AMIR MOUSAVI

Machine Learning Research Engineer | Applied AI & Scientific Software Engineer

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PROFESSIONAL SUMMARY

Machine Learning Research Engineer and PhD candidate in Neuroscience with a multidisciplinary foundation in electrical engineering, applied mathematics, statistics, optimization, and software engineering. Experience has evolved from embedded systems and backend development to machine learning and deep learning for medical imaging, computer vision, and clinical data analysis. Combines mathematical reasoning, rigorous research methods, and software engineering to formulate complex problems, develop and validate AI/ML solutions, and translate them into reliable software systems. Particularly motivated by real-world challenges that require working across disciplines and connecting research, mathematics, AI, and engineering.

CORE EXPERTISE

Machine Learning & AI	Machine learning, deep learning, computer vision, statistical learning, Bayesian methods, graph-based learning, generative AI, optimization
ML & Scientific Computing	Python, NumPy, pandas, scikit-learn, XGBoost, TensorFlow, Keras, PyTorch, model evaluation, cross-validation, visualization
Software Engineering	Python, JavaScript, FastAPI, Django, Flask, Node.js, Express, Next.js, React, REST APIs
Applied AI & RAG	RAG, embeddings, semantic chunking, vector search, keyword and hybrid retrieval, grounded question answering
Data & Infrastructure	PostgreSQL, MongoDB, Redis, OpenSearch, Docker, Nginx, Linux, Git, GitHub Actions, CI/CD

SELECTED PROJECTS & ENGINEERING EXPERIENCE

DocuMind — Scientific Document Intelligence & RAG Platform

Applied AI, RAG & Software Engineering | 2026

- Architected and developed a self-hosted, multi-service platform for organizing scientific PDFs, retrieving evidence, and answering grounded questions with citations.
- Implemented document extraction, semantic chunking, embeddings, and vector, keyword, and hybrid retrieval using OpenSearch and local model inference.
- Designed provenance-aware asynchronous processing pipelines integrating Python services, Node.js, Express, Next.js, PostgreSQL, Redis, and REST APIs.
- Implemented authentication, Docker/Nginx deployment, and CI/CD workflows for reliable multi-service operation.

PaperMind — Scientific PDF Processing Services

Scientific Software & Document Processing | 2026

- Developed Python and FastAPI services that transform scientific publications into structured, machine-readable packages containing metadata, hierarchical text, references, figures, tables, formulas, provenance, and quality information.
- Designed reusable API and command-line workflows for document validation, indexing, retrieval, and RAG pipelines, and integrated the services as the document-processing layer of DocuMind.

Baxter Chess-Playing Robot — Vision-Based Robotic Chess System

Computer Vision, Robotics & Software Engineering | 2026

- Developed a robotic chess system integrating Kinect-based computer vision, chess-state reasoning, Stockfish, FastAPI, ROS, calibration, and motion control.
- Implemented the workflow for detecting human chess moves, updating the board state, generating the robot's response, and commanding a Baxter robot to physically manipulate chess pieces.

Brain Ischemic Stroke Lesion Segmentation

Deep Learning & Medical Imaging | 2025–2026

- Developed deep-learning architectures for automated detection and segmentation of ischemic-stroke lesions from T1-weighted brain MRI.
- Designed and evaluated segmentation pipelines incorporating convolutional, residual, and attention-based components, covering preprocessing, training, validation, and quantitative evaluation.
- Research resulted in RAX-NET and VRXU-net, released as scientific preprints.

Lumbar Spine Degeneration Classification

Deep Learning & Medical Imaging | 2024

- Developed a deep-learning approach for automated classification of lumbar-spine degeneration using YOLOv8 and DeepScoreNet.
- Built and evaluated a computer-vision pipeline for identifying relevant spinal structures and classifying degenerative conditions.
- Research resulted in a manuscript released as a medRxiv preprint.

Synthetic Face Generation — Eyes-to-Face Inpainting

Computer Vision, Generative Modeling & Biometrics | IEEE IJCB 2023

- Developed an eyes-to-face inpainting approach that generates synthetic facial images by combining eye regions from different individuals and reconstructing faces.
- Evaluated synthetic datasets for age and gender estimation and face-recognition model training, including validation across four benchmark face-recognition datasets.
- Investigated demographic biases and the potential of synthetic data to improve representation of groups with limited samples in biometric datasets.

COVID-19 Lung & Lesion Segmentation

Deep Learning, CT Imaging & Web Deployment | 2019 - 2023

- Developed a deep-learning pipeline for automated lung and COVID-19 lesion segmentation from chest CT scans using clinical imaging data collected from hospitals.
- Built and trained segmentation models to quantify affected lung regions and support analysis of COVID-19-related lung damage.
- Integrated preprocessing, inference, and visualization into a web-based application designed for practical use by healthcare professionals.

Real-Time Face-Mask Detection

Edge AI & Computer Vision | 2019 - 2023

- Developed a real-time deep-learning system for detecting face-mask usage using a Raspberry Pi camera.
- Implemented image acquisition, preprocessing, inference, and visualization for deployment on resource-constrained edge hardware.

Backend Developer - Maralhost

Isfahan, Iran | May 2021- Oct 2022

- Developed and maintained Django-based backend services and API microservices for web and server-management applications, including relational data models and server-side business logic.
- Designed and implemented authentication and authorization mechanisms, including Single Sign-On (SSO), to provide secure access across multiple services and applications.
- Developed tools for web vulnerability assessment and supported Linux-based deployment, debugging, database-backed services, and collaborative Git workflows.

Programmer & Electronics Designer - Mobarakeh Steel Company

Isfahan, Iran | Oct 2017 - Oct 2019

- Analyzed and redesigned an existing electronic control board for rolling-equipment speed regulation, integrating a microcontroller-based control system.
- Developed and implemented control algorithms and evaluated system performance, behavior, and reliability under operational conditions.

EDUCATION

PhD Candidate in Neuroscience — Université de Picardie Jules Verne (UPJV)

Amiens, France | 2023 - Expected Sep 2026

- Develop and validate machine-learning models, including Support Vector Machines (SVM), regression, and ensemble methods, using brain-connectivity and clinical data from approximately 400 stroke patients to model cognitive and motor outcomes, including executive function, processing speed, and motor function.
- Extract and analyze brain-connectivity and graph-derived features, identifying statistically significant voxel-level patterns associated with specific cognitive and motor functions and improving upon previous findings.
- Develop computational simulations of brain functional dynamics using the Kuramoto model to investigate network synchronization and connectivity patterns.
- Develop Graph Neural Network (GNN) approaches using structural brain-connectivity data to investigate relationships between brain-network organization and post-stroke functional outcomes.
- Build reproducible Python research pipelines covering data preprocessing, feature engineering, hyperparameter optimization, cross-validation, statistical analysis, visualization, and model interpretation.
- Apply statistical, Bayesian, graph-based, and optimization methods while collaborating with neuroscientists, healthcare researchers, statisticians, and engineers to translate multidisciplinary research questions into computational experiments.
- Presented research at **NetSci 2025** in Maastricht and the **OHBM 2026** Annual Meeting in Bordeaux; manuscript currently under review.

MSc in Electrical Engineering — Telecommunications — Shiraz University of Technology

Shiraz, Iran | 2019 - 2023

- Thesis: Quantification of damaged lung tissue due to COVID-19 using computed tomography images.

BEng in Electrical Engineering — Telecommunications — University of Isfahan

Isfahan, Iran | 2015 - 2019

- Final project: Persian-language e-commerce sentiment analysis with positive/negative classification and negative-term extraction.

SELECTED PUBLICATIONS

Mousavi Mobarakeh, S. A. *VRXU-net: A Deep Learning Approach for Brain Ischemic Stroke Lesion Detection and Segmentation in T1W MRI.* arXiv, 2026. doi:10.48550/arXiv.2605.21633

Mousavi, A. *RAX-NET: Residual Attention Xception Network for Brain Ischemic Stroke Segmentation in T1-Weighted MRI.* medRxiv, 2025. doi:10.1101/2025.09.26.25336733

Mousavi, A. *Lumbar Spine Degenerative Classification Using YOLO v8 and DeepScoreNet.* medRxiv, 2024. doi:10.1101/2024.12.06.24318595

Mousavi Mobarakeh, S. A.; Kazemi, K.; Aarabi, A.; Danyal, H. *Empowering Medical Imaging with Artificial Intelligence: A Review of Machine Learning Approaches for the Detection, and Segmentation of COVID-19 Using Radiographic and Tomographic Images.* arXiv, 2024. doi:10.48550/arXiv.2401.07020

Hassanpour, A.; **Mousavi Mobarakeh, S. A.**; Etefaghi Daryani, A.; Ramachandra, R.; Yang, B. *Synthetic Face Generation Through Eyes-to-Face Inpainting.* IEEE IJCB, 2023, pp. 1-8. doi:10.1109/IJCB57857.2023.10449216

Full publication list: Google Scholar

TEACHING & ADDITIONAL INFORMATION

Teaching: Teaching Assistant for approximately 60 hours of university-level Python and machine-learning instruction, laboratory support, and student guidance (2025-2026).

Languages: Persian/Farsi (native); English (professional proficiency); French (B1 listening, A2 speaking, A1 writing).